Lecture 11

The Jucys–Murphy spectrum

Objectives. *Lecture 10 gave each irreducible representation of S_n a Gelfand–Tsetlin basis, whose vectors are separated by their weights $\alpha = (a_1, \dots, a_n)$. Which weights occur? Every weight is a* content vector*, cut out by three conditions: the proof studies s_i on the span of v and $s_i v$, with an invariant inner product, and uses one Coxeter relation. Content vectors are the contents of standard Young tableaux. Lecture 12 proves that every content vector is a weight.*

By the end of this lecture you will be able to:

- Prove that a GZ vector v of weight α has $a_{i+1} \neq a_i$, that $s_i v = \pm v$ when $a_{i+1} = a_i \pm 1$, and that otherwise $s_i v - (a_{i+1} - a_i)^{-1} v$ is a GZ vector of weight $s_i \alpha$; compute s_i on the two vectors.
- Use the Coxeter relation $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ to show that no weight has three consecutive entries a, $a \pm 1$, a.
- Prove that every weight of S_n is a content vector, and decide whether a given vector is one.
- Prove that $T \mapsto (c_1(T), \dots, c_n(T))$ maps the standard Young tableaux with n boxes bijectively onto the content vectors of length n.

11.1 Two-dimensional analysis

Throughout, $V \in \mathrm{Irr}(S_n)$ has its lines V_γ (Theorem 10.5), and a GZ vector v has the weight $\alpha(v) = (a_1, \dots, a_n)$ of Definition 10.7. Fix on V an invariant inner product (Theorem 5.9(iii)), for which each $\rho(g)$ is unitary.

Lemma 11.1 (GZ vectors and the inner product)**.**

(i) Each s_i and each X_k acts on V by a self-adjoint operator, so the entries of every weight are real.

(ii) The lines V_γ are pairwise orthogonal.

(iii) If $w \in V$ is non-zero and $X_k w = b_k w$ for every k, then w is a GZ vector, of weight $(b_1, \dots, b_n)$.

Proof. (i) A transposition t has $t^2 = e$, so

$$\rho(t)^\dagger = \rho(t)^{-1} = \rho(t).$$

Each X_k is a sum of transpositions. Self-adjoint operators have real eigenvalues.

(ii) Two different paths have different weights (Corollary 10.13), so some X_k acts on their lines by different scalars. Eigenvectors of a self-adjoint operator for different eigenvalues are orthogonal.

(iii) Write $w = \sum_\gamma w_\gamma$ with $w_\gamma \in V_\gamma$. Each $(X_k - b_k)w_\gamma$ lies in V_γ, and the sum of the lines is direct, so $(X_k - b_k)w_\gamma = 0$ for every γ and k. So each γ with $w_\gamma \neq 0$ has weight $(b_1, \dots, b_n)$, and by Corollary 10.13 there is only one. ∎

Theorem 11.2 (The local structure). *Let v be a GZ vector of weight $\alpha = (a_1, \dots, a_n)$, and $1 \le i \le n-1$. Then $a_{i+1} \neq a_i$. Put $r = a_{i+1} - a_i$ and*

$$w = s_i v - r^{-1} v.$$

(i) If $r = \pm 1$, then $s_i v = rv$.

(ii) If $r \neq \pm 1$, then w is a GZ vector of weight

$$s_i\alpha = (a_1, \dots, a_{i-1}, a_{i+1}, a_i, a_{i+2}, \dots, a_n),$$

orthogonal to v, with $\|w\|^2 = (1 - r^{-2}) \|v\|^2$, and

$$s_i v = r^{-1} v + w, \qquad s_i w = (1 - r^{-2})\, v - r^{-1} w.$$

Proof. $r \neq 0$. Suppose $a_{i+1} = a_i$, and call it a. By Corollary 10.17,

$$(X_i - a)s_i v = -v, \qquad (X_i - a)v = 0.$$

As a is real, $X_i - a$ is self-adjoint (Lemma 11.1(i)), and

$$\|v\|^2 = -\langle v, (X_i - a)s_i v\rangle = -\langle (X_i - a)v, s_i v\rangle = 0,$$

a contradiction.

A weight vector. By Corollary 10.17, and as $a_i + r = a_{i+1}$,

$$\begin{aligned} X_i w &= X_i s_i v - r^{-1} a_i v \\ &= a_{i+1} s_i v - v - r^{-1} a_i v \\ &= a_{i+1} s_i v - r^{-1} a_{i+1} v = a_{i+1} w. \end{aligned}$$

In the same way, from the second relation of Corollary 10.17,

$$X_{i+1} w = a_i w, \qquad X_k w = a_k w \quad (k \neq i, i+1).$$

So if $w \neq 0$, it is a GZ vector of weight $s_i\alpha$ (Lemma 11.1(iii)). As $r \neq 0$, $s_i\alpha \neq \alpha$, and w is orthogonal to v (Lemma 11.1(ii)).

The norm. So $s_i v = r^{-1} v + w$ with w orthogonal to v, and s_i is unitary:

$$\|v\|^2 = \|s_i v\|^2 = r^{-2} \|v\|^2 + \|w\|^2 .$$

So $\|w\|^2 = (1 - r^{-2}) \|v\|^2$.

(i) If $r = \pm 1$, then $w = 0$, and $s_i v = r^{-1} v$, which is rv as $r^2 = 1$.

(ii) If $r \neq \pm 1$, then $w \neq 0$, and only $s_i w$ remains. As $s_i^2 = e$,

$$s_i w = v - r^{-1} s_i v = (1 - r^{-2})\, v - r^{-1} w.$$ ∎

Lecture 12 normalises v and w, so that the coefficient of $w/\|w\|$ in $s_i v$ becomes $\sqrt{1 - r^{-2}}$: Young's orthogonal form.

Example 11.3 (E again)**.** In § 10.6, $u = (1, 1, -2)$ and $f_1 = (1, -1, 0)$ are GZ vectors of E, of weights $(0, 1, -1)$ and $(0, -1, 1)$, and the inner product of $\mathbb{C}^3$ is invariant. For $i = 1$, r is 1 for u and -1 for f_1, and indeed

$$s_1 u = u, \qquad s_1 f_1 = -f_1.$$

For $i = 2$ and u, $r = -2$, and by Lecture 10's self-check

$$w = s_2 u + \frac{1}{2} u = \left(-\frac{1}{2} u + \frac{3}{2} f_1\right) + \frac{1}{2} u = \frac{3}{2} f_1,$$

a GZ vector of the swapped weight $(0, -1, 1)$, as the local theorem predicts. Its norm is

$$\|w\|^2 = \frac{9}{4} \cdot 2 = \left(1 - \frac{1}{4}\right) \cdot 6 = (1 - r^{-2}) \|u\|^2 .$$

Example 11.4 (Guard: the relations alone)**.** The local theorem's $r = \pm 1$ case used the inner product. The relations alone do not suffice. On $\mathbb{C}^2$ with basis v, w, let

	v	w
X_i	av	$(a+1)w$
X_{i+1}	$(a+1)v$	aw
s_i	$v + w$	$-w$

Then

$$s_i^2 = \mathbb{1}, \qquad s_i X_i + 1 = X_{i+1} s_i, \qquad X_i s_i + 1 = s_i X_{i+1}.$$

Here $a_{i+1} = a_i + 1$ for v, but $s_i v \neq v$. An invariant line is an eigenline $\mathbb{C}v$ or $\mathbb{C}w$ of X_i, and $s_i v \notin \mathbb{C}v$: the module does not split. Indeed, an inner product making X_i and s_i self-adjoint would make v and w orthogonal, and then

$$\langle s_i v, w \rangle = \|w\|^2 , \qquad \langle v, s_i w \rangle = 0.$$

In characteristic dividing $n!$, Lecture 10's construction does not start (Example 5.10).

11.2 The braid exclusion

Definition 11.5 (Admissible transpositions)**.** For $\alpha = (a_1, \ldots, a_n) \in \mathbb{C}^n$ and $1 \le i \le n-1$, $s_i \alpha$ exchanges a_i and a_{i+1}. The transposition s_i is *admissible* for α if $a_{i+1} \neq a_i \pm 1$.

By Theorem 11.2(ii), if α is the weight of a GZ vector of V and s_i is admissible for α, then $s_i \alpha$ is a weight of the same V.

Lemma 11.6 (The braid exclusion)**.** *Let $n \ge 3$ and $1 \le i \le n-2$. No weight $\alpha \in \mathrm{Spec}(n)$ has*

$$a_{i+2} = a_i, \qquad a_{i+1} = a_i \pm 1.$$

Proof. Suppose a GZ vector v has $a_{i+1} = a_i + 1$ and $a_{i+2} = a_i$, so that $a_{i+2} = a_{i+1} - 1$. By Theorem 11.2(i),

$$s_i v = v, \qquad s_{i+1} v = -v.$$

So

$$s_i s_{i+1} s_i v = s_i s_{i+1} v = -s_i v = -v,$$

$$s_{i+1} s_i s_{i+1} v = -s_{i+1} s_i v = -s_{i+1} v = v.$$

The Coxeter relation of § 10.5 makes these equal, so $v = -v$, and $v = 0$, a contradiction. If $a_{i+1} = a_i - 1$, exchange the signs. ∎

Example 11.7 (The vector $(0,1,0)$)**.** The vector $(0,1,0)$ passes every test of Theorem 11.2: no two neighbours are equal, and r is 1 for $i = 1$ and -1 for $i = 2$. A GZ vector of this weight would be fixed by s_1 and negated by s_2, which the braid relation forbids. So $(0,1,0) \notin \mathrm{Spec}(3)$.

Lemma 11.8 (Moving an entry)**.** *Let α be the weight of a GZ vector of V, and $p \le q$.*

(i) If $a_j \neq a_q \pm 1$ for $p \le j < q$, then moving the entry a_q to place p, and $a_p, \dots, a_{q-1}$ one place to the right, gives the weight of a GZ vector of V.

(ii) If $a_j \neq a_p \pm 1$ for $p < j \le q$, then moving a_p to place q, and $a_{p+1}, \dots, a_q$ one place to the left, does too.

Proof. (i) Apply s_{q-1}, then s_{q-2}, and so on down to s_p. Each exchanges the moving entry a_q with its left neighbour a_j, $p \le j < q$, and is admissible, as $a_j \neq a_q \pm 1$; so each step keeps a weight of V (Theorem 11.2(ii)).

(ii) The same, with s_p, then s_{p+1}, up to s_{q-1}. ∎

Lemma 11.9 (Truncation)**.**

(i) Every weight has $a_1 = 0$.

(ii) If $n \ge 2$ and $(a_1, \dots, a_n) \in \mathrm{Spec}(n)$, then $(a_1, \dots, a_{n-1}) \in \mathrm{Spec}(n-1)$.

(iii) The weights of S_1 and S_2 are

$$\mathrm{Spec}(1) = \{(0)\}, \qquad \mathrm{Spec}(2) = \{(0,1),\ (0,-1)\}.$$

Proof. (i) $X_1 = 0$.

(ii) Let v be a GZ vector of V, in the line of $\gamma = (U_1 \to \cdots \to U_n)$. Then v lies in $W_{n-1} \cong U_{n-1}$ (Theorem 10.5(i)). An isomorphism $\varphi : W_{n-1} \to U_{n-1}$ of S_{n-1}-representations commutes with $X_1, \dots, X_{n-1}$, which lie in $\mathbb{C}[S_{n-1}]$. So $X_k \varphi(v) = a_k \varphi(v)$ for $k \le n-1$, and $\varphi(v)$ is a GZ vector of U_{n-1} of weight $(a_1, \dots, a_{n-1})$, by Lemma 11.1(iii) for S_{n-1}.

(iii) S_1 has only $\mathbf{1}$. The irreducibles of S_2 are the lines $\mathbf{1}$ and sgn, and $X_2 = (1\,2)$ acts on them by 1 and -1. ∎

11.3 $\mathrm{Spec}(n) \subseteq \mathrm{Cont}(n)$

Definition 11.10 (Content vectors)**.** A vector $\alpha = (a_1, \dots, a_n) \in \mathbb{C}^n$ is a *content vector* if

(C1) $a_1 = 0$;

(C2) for $1 < q \le n$, $a_q - 1$ or $a_q + 1$ is one of $a_1, \dots, a_{q-1}$;

(C3) if $a_p = a_q = a$ with $p < q$, then $a - 1$ and $a + 1$ are both among $a_{p+1}, \dots, a_{q-1}$.

$\mathrm{Cont}(n)$ is the set of content vectors of length n.

By (C1) and (C2), every entry of a content vector is an integer. The conditions on a_q involve only $a_1, \dots, a_q$, so for $n \ge 2$ dropping the last entry of a content vector of length n gives one of length $n-1$. The name comes from § 11.4.

Example 11.11 (Cont(3))**.** By (C2), $a_2 = \pm 1$. Let $a_2 = 1$. By (C2), $a_3 - 1$ or $a_3 + 1$ is 0 or 1, so

$$a_3 \in \{-1, 0, 1, 2\}.$$

(C3) removes 1, as nothing lies between the two 1s, and 0, as -1 does not lie between the two 0s. The case $a_2 = -1$ is the mirror image, so

$$\mathrm{Cont}(3) = \{(0,1,2),\ (0,1,-1),\ (0,-1,1),\ (0,-1,-2)\}.$$

Theorem 11.12 (Weights are content vectors). *For every $n \geq 1$, $\mathrm{Spec}(n) \subseteq \mathrm{Cont}(n)$. In particular, every weight is a vector of integers.*

Proof. By induction on n. The sets $\mathrm{Spec}(1)$ and $\mathrm{Spec}(2)$ of Lemma 11.9(iii) are $\mathrm{Cont}(1)$ and $\mathrm{Cont}(2)$, by (C1) and (C2). Let $n \geq 3$, and let $\alpha \in \mathrm{Spec}(n)$ be the weight of a GZ vector of V. By Lemma 11.9(ii) and induction, $(a_1, \dots, a_{n-1}) \in \mathrm{Cont}(n-1)$; so (C1) holds, and (C2) and (C3) hold for $q < n$. Let $a = a_n$.

(C2) for $q = n$. Suppose neither $a-1$ nor $a+1$ is among $a_1, \dots, a_{n-1}$. Then s_{n-1} is admissible for α, so $(a_1, \dots, a_{n-2}, a, a_{n-1}) \in \mathrm{Spec}(n)$, and by Lemma 11.9(ii) and induction

$$(a_1, \dots, a_{n-2}, a) \in \mathrm{Spec}(n-1) \subseteq \mathrm{Cont}(n-1).$$

Its last entry a stands in place $n-1 \geq 2$, so by (C2) $a-1$ or $a+1$ is among $a_1, \dots, a_{n-2}$: a contradiction.

(C3) for $q = n$: the set-up. Let $a_p = a$ with $p < n$, and take p largest, so that a is not among $a_{p+1}, \dots, a_{n-1}$; a smaller p only lengthens the range in (C3). Suppose $a-1$ is not among them either; the case of $a+1$ is the same with the signs exchanged.

At most one $a+1$. If a_j and $a_{j'}$, $p < j < j' < n$, both equal $a+1$, then (C3) for $(a_1, \dots, a_{n-1})$ puts a between them, against the choice of p. So every a_j with $p < j < n$, except at most one equal to $a+1$, has $a_j \neq a$ and $a_j \neq a \pm 1$.

No $a+1$. Move a_n to place $p+1$ (Lemma 11.8(i)). The result is a weight with two equal neighbours a, a in places p and $p+1$, against Theorem 11.2.

One $a+1$, at j. Move a_p to place $j-1$ and then a_n to place $j+1$ (Lemma 11.8(ii), (i)). The result is a weight with the consecutive entries a, $a+1$, a, against Lemma 11.6. ∎

Lecture 12 proves the converse, $\mathrm{Cont}(n) \subseteq \mathrm{Spec}(n)$, by counting.

11.4 Content vectors and standard tableaux

Definition 11.13 (Young diagrams and standard tableaux). Let $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash n$ (Definition 2.8). Its *Young diagram* is the set of boxes $b = (i, j)$ with $1 \leq i \leq \ell$ and $1 \leq j \leq \lambda_i$, drawn the English way: rows from top to bottom, each starting in the first column. λ is its *shape*. The *content* of $b = (i, j)$ is $c(b) = j - i$; the boxes of one content form a *diagonal*. A *standard Young tableau* T of shape λ fills the boxes with $1, \dots, n$, each once, increasing along every row and down every column. $\mathrm{SYT}(\lambda)$ is the set of them, $\mathrm{SYT}(n)$ the union over $\lambda \vdash n$, and $c_k(T)$ is the content of the box holding k.

For instance, the tableau

$$T = \begin{array}{|c|c|c|}\hline 1 & 2 & 4 \\ \hline 3 & 5 \\ \cline{1-2}\end{array} \qquad \text{has} \qquad (c_1(T), \dots, c_5(T)) = (0, 1, -1, 2, 0).$$

Three facts, for T of shape λ:

(a) If a box b' lies weakly above and to the left of another box b, the entry of T in b' is smaller: go along the row of b' to the column of b, then down. So the entries increase down each diagonal.

(b) The contents of the boxes of λ are the integers from $1-\ell$ to $\lambda_1 - 1$: the first row and the first column give them all.

(c) A box whose addition leaves a diagram, an *addable* box, is $(i, \lambda_i + 1)$ for some $i \le \ell + 1$, with $\lambda_{\ell+1} = 0$. Its content $\lambda_i + 1 - i$ strictly decreases with i, so different addable boxes have different contents.

Proposition 11.14 (Content vectors and tableaux)**.** *The map $T \mapsto (c_1(T), \dots, c_n(T))$ is a bijection from* $\mathrm{SYT}(n)$ *onto* $\mathrm{Cont}(n)$.

Proof. Into $\mathrm{Cont}(n)$. Let $a_k = c_k(T)$. (C1): 1 is in $(1,1)$. (C2): if $k > 1$ is in $b = (i,j)$, then $(i, j-1)$ or $(i-1, j)$ is a box, with a smaller entry by increase towards the lower right, and content $a_k - 1$ or $a_k + 1$.

(C3): if $a_p = a_q$ with $p < q$, then p and q lie on one diagonal, q lower, in $b = (i,j)$ say, with $i, j \ge 2$. The boxes $(i-1, j)$ and $(i, j-1)$, of contents $a_q + 1$ and $a_q - 1$, hold entries between p and q, by the same increase.

One-to-one. By increase down diagonals, the diagonal of content h holds the k with $a_k = h$, in increasing order from the top. The boxes of a diagonal run unbroken from its top box, so the vector places each k.

Onto. By induction on n; $n = 1$ is clear. Let $\alpha \in \mathrm{Cont}(n)$, $n \ge 2$. By induction, $(a_1, \dots, a_{n-1})$ is the vector of some $T' \in \mathrm{SYT}(\mu)$, $\mu \vdash n - 1$. It suffices to put n into an addable box of μ of content $a = a_n$.

If a is among $a_1, \dots, a_{n-1}$, let p be its last place and $b = (i,j)$ the box of p, the lowest on its increasing diagonal.

By (C3), the diagonal of content $a + 1$ has a box with an entry larger than p. By increase towards the lower right, that box is not above b, so it is $(i + m, j + 1 + m)$ for some $m \ge 0$, and $(i, j+1) \in \mu$. In the same way $(i+1, j) \in \mu$.

So $(i+1, j+1)$ is addable, of content a.

If not, then $a \ne 0$, by (C1). Let $a > 0$. By (C2), $a - 1$ or $a + 1$ is a content of μ.

The content interval of μ runs from $1 - \ell(\mu)$ to $\mu_1 - 1$. As $a > 0$ is not among them, $a \ge \mu_1$; so $a - 1$ is one, and $a = \mu_1$.

The box $(1, \mu_1 + 1)$ is addable, of content a.

If $a < 0$, likewise $a = -\ell(\mu)$, and $(\ell(\mu) + 1, 1)$ is addable. ∎

The proposition is combinatorial: it labels no GZ vector until Lecture 12 proves $\mathrm{Spec}(n) = \mathrm{Cont}(n)$.

11.5 Worked example

What are the weights of S_3? By Theorem 11.12 they lie in $\mathrm{Cont}(3)$, which has four elements (Example 11.11). S_3 has four paths of length 3, one each to $\mathbf{1}$ and sgn and two to E, with different weights (Corollary 10.13). So $\mathrm{Spec}(3) = \mathrm{Cont}(3)$.

On $\mathbf{1}$ each X_k acts by $k - 1$, and on sgn by $1 - k$: their weights $(0, 1, 2)$ and $(0, -1, -2)$ are the vectors of the one-row and the one-column tableaux. E has the other two, $(0, 1, -1)$ for u and $(0, -1, 1)$ for f_1 (Example 11.3), the vectors of

$$\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array} \quad \text{and} \quad \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}.$$

And $(0, 1, 0)$ is not a weight (Example 11.7).

Self-check. Which of $(0,1,-1,0)$ and $(0,1,0,-1)$ are content vectors, and of which tableaux? *Answer:* in $(0,1,-1,0)$, both 1 and -1 lie between the two 0s, and (C2) holds; it is the vector of

$$\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array},$$

of shape $(2,2)$, and the weight of $\rho_E \circ \pi$ in Example 10.9. In $(0,1,0,-1)$, -1 does not lie between the two 0s, so it fails (C3) and is not a weight.

Summary.

- *The main results:* the local structure of s_i (Theorem 11.2); the braid exclusion (Lemma 11.6); every weight is a content vector (Theorem 11.12); content vectors are the contents of standard tableaux (Proposition 11.14).
- *Handed on:* the local moves, $\mathrm{Spec}(n) \subseteq \mathrm{Cont}(n)$ and $\mathrm{Cont}(n) \leftrightarrow \mathrm{SYT}(n)$ to Lecture 12.
- *Next:* Lecture 12 proves $\mathrm{Spec}(n) = \mathrm{Cont}(n)$, classifies the irreducibles by partitions, and writes Young's orthogonal form.

References

Vershik and Okounkov, *A new approach to the representation theory of the symmetric groups. II* (arXiv:math/0503040): the algebra $H(2)$, pp. 14–15; Prop. 4.1, p. 16; Thm 5.1 and Lemma 5.2, pp. 17–19; Prop. 5.3, p. 20. Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Def. 3.1.8 and Thm 3.1.10, pp. 86–89; Prop. 3.3.3 and Lemmas 3.3.4–3.3.6, pp. 103–106.[1]

[1] They write $\mathrm{Tab}(\lambda)$ for $\mathrm{SYT}(\lambda)$ and $C(T)$ for the vector of T. Both sources argue for Theorem 11.2(i) in the span of v and $s_i v$ alone; Example 11.4 shows why this course uses the inner product there.